

How Well-Calibrated Are Driving World Models? A Benchmark for Predictive Uncertainty

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Abstract

World models for autonomous driving are increasingly used for planning and simulation, yet their predictive uncertainty remains poorly understood. While these models can generate realistic future states, overconfident or poorly calibrated predictions can lead to catastrophic downstream planning failures. To address this gap, we introduce CalibraDrive, the first systematic benchmark for evaluating the calibration of driving world models. We evaluate representative architectures, including 3D occupancy (OccWorld) and video generation (Vista) models, across spatial dimensions, temporal horizons, and scenario difficulties on the nuScenes dataset. Our findings indicate that all tested models are significantly overconfident ($ECE > [TODO: 0.05]$), with calibration degrading further over longer prediction horizons and at greater distances from the ego vehicle. Furthermore, we demonstrate that simple post-hoc recalibration methods can reduce the ECE by over $[TODO: 50\%]$. Importantly, integrating calibrated uncertainty measurably improves downstream planning safety, highlighting the critical need for uncertainty-aware world modeling in autonomous driving.

1. Introduction

World models for autonomous driving are transforming the landscape of vehicle planning and simulation [3, 5, 11, 13, 16]. By learning a predictive distribution over future driving states conditioned on current context and ego actions, these generative models offer an elegant, scalable alternative to hand-crafted simulation environments. However, while world models have demonstrated impressive capability in generating realistic multi-modal futures, a critical question remains largely unanswered: do we know when these models are wrong?

When a driving world model makes an overconfident but incorrect prediction about the future trajectory of a pedestrian or the occupancy of an intersection, downstream planning algorithms that rely on these predictions can make unsafe, potentially catastrophic decisions. Predictive uncertainty is essential for safe autonomous driving [7]. In practice, an autonomous vehicle should act cautiously when its predictive model is uncertain, prioritizing safety over progress. But if the world model itself produces poorly calibrated probabilities—predicting high confidence for incorrect futures—the planner is effectively flying blind.

Calibration has been extensively studied in standard deep learning tasks like image classification and object detection [4], yet it remains largely unexplored for generative driving world models. Existing literature on uncertainty in autonomous driving primarily focuses on perception-level uncertainty or traditional prediction models, lacking a systematic evaluation of modern large-scale world models. While recent methods like C3 [7] and RELIOCC have proposed specific calibration techniques for certain prediction tasks, there currently exists no standardized benchmark for evaluating and comparing the calibration of diverse driving world model architectures.

In this paper, we present CalibraDrive, the first systematic benchmark for evaluating the predictive uncertainty and calibration of driving world models. Our framework spans diverse representative architectures, evaluating both 3D occupancy forecasting (OccWorld [16]) and video generation (Vista [3]) paradigms. Through extensive spatial-temporal analysis, we reveal where and when these models fail, demonstrating that confidence is poorly aligned with accuracy, particularly at longer prediction horizons and in hard driving scenarios.

In summary, our key contributions are:

1. CalibraDrive Benchmark: We introduce the first comprehensive framework for evaluating calibration

[TODO: Include Teaser Figure]

Figure 1. The CalibraDrive benchmark framework. We evaluate the calibration of driving world models (e.g., occupancy and video generation) across spatial and temporal dimensions. Poorly calibrated models lead to unsafe planning, while calibrated uncertainty significantly improves downstream safety.

in generative driving world models across different architectures (occupancy and video).

2. Spatial-Temporal Calibration Analysis: We provide an in-depth analysis of where and when driving world models are miscalibrated, showing degradation over prediction horizons and spatial distances.
3. Recalibration and Downstream Safety: We demonstrate that simple post-hoc recalibration techniques not only significantly improve calibration metrics (e.g., reducing Expected Calibration Error) but also measurably enhance the safety of downstream planning algorithms.

2. Related Work

Driving World Models. The success of large-scale generative models has spurred significant interest in developing world models for autonomous driving. These models generally predict future states conditioned on historical context and ego-vehicle actions, enabling both closed-loop simulation and model-based planning. OccWorld [16] predicts future 3D occupancy and ego trajectories autoregressively. Video-based world models such as Vista [3], GAIA-1 [5], and DriveDreamer [11] formulate the problem as future video generation. Other notable works include UniSim [13], GenAD [15], Copilot4D [14], and Drive-WM [12]. While these models achieve impressive fidelity in predicting future states, their predictive confidence and uncertainty have not been rigorously evaluated.

Calibration in Deep Learning. Calibration measures the alignment between a model’s predicted confidence and its actual empirical accuracy. Deep neural networks, particularly large modern architectures, are often notoriously overconfident [4]. Various methods have been proposed to improve calibration, ranging from post-hoc scaling methods like Platt scaling [8] and temperature scaling [4] to distribution-free guarantees using conformal prediction [9, 10]. Other prominent approaches include Bayesian Neural Networks, MC Dropout [2], and Deep Ensembles [6]. While these techniques are well-established for classification and regression, adapting and evaluating them for high-dimensional, generative driving world models remains

an open challenge.

Uncertainty in Autonomous Driving. Uncertainty estimation is critical for safe autonomous driving. While extensive research has addressed uncertainty in 3D object detection, semantic segmentation, and traditional trajectory prediction, exploring uncertainty in comprehensive driving world models is a nascent field. The most closely related work to ours is C3 [7], which explores calibration for conformal trajectory prediction, and RELIOCC, which targets occupancy calibration. However, these works primarily propose specific methodologies for specific tasks. In contrast, CalibraDrive provides a standardized benchmark and systemic evaluation across fundamentally different world model architectures (e.g., 3D occupancy vs. video generation), temporal horizons, and planning settings.

3. Methodology

3.1. Problem Formulation

We define a driving world model W as a probabilistic model that takes historical context c (e.g., past sensor data, maps) and future ego actions a to produce a distribution over future states. Let $Y \in \mathcal{Y}$ be the true future state. The model produces a predictive distribution $P_\theta(Y|c, a)$, from which we can extract a confidence score $\hat{p}$ for a specific prediction $\hat{Y}$.

A model is perfectly calibrated if the confidence $\hat{p}$ matches the empirical probability of the prediction being correct. Formally:

$$\mathbb{P}(Y = \hat{Y} \mid \hat{p} = p) = p, \quad \forall p \in [0, 1]. \quad (1)$$

For 3D occupancy models, we evaluate binary calibration per voxel, where $\hat{p}$ is the predicted probability of the voxel being occupied. For video generation, we assess the predictive variance and probability of semantic classes per pixel.

3.2. Calibration Metrics

To quantify calibration, we employ standard metrics adapted for the spatial-temporal nature of driving predictions.

- **Expected Calibration Error (ECE):** We partition the confidence scores into B bins. The ECE is the weighted average of the absolute difference between accuracy and confidence across bins:

$$\text{ECE} = \sum_{b=1}^B \frac{|B_b|}{N} |\text{acc}(B_b) - \text{conf}(B_b)|, \quad (2)$$

where N is the total number of predictions, and B_b represents the set of predictions falling into bin b .

- Maximum Calibration Error (MCE): The maximum discrepancy across all bins: $\max_b |\text{acc}(B_b) - \text{conf}(B_b)|$.
- Brier Score: A strictly proper scoring rule measuring the mean squared error of probability predictions: $\frac{1}{N} \sum_{i=1}^N (p_i - y_i)^2$.
- AUROC & Reliability Diagrams: We use AUROC to evaluate discriminative ability, independent of calibration. Reliability diagrams visually plot accuracy versus confidence.

3.3. Spatial-Temporal Analysis

Driving predictions are inherently spatial and temporal. A prediction error directly in front of the ego vehicle is significantly more critical than one 100 meters away. Thus, CalibraDrive stratifies evaluation across several axes:

- Spatial Distance: We evaluate calibration in discrete radial distance bins from the ego vehicle: $[0 - 10m, 10 - 30m, 30 - 50m, 50 - 100m]$.
- Temporal Horizon: We analyze calibration across prediction horizons from 0.5s up to 3.0s into the future.
- Scenario Difficulty: We classify scenarios as easy, medium, or hard based on agent density, road curvature, and average time-to-collision (TTC).
- Dynamic vs. Static: We separately evaluate calibration for static background elements versus dynamic traffic participants.

3.4. Post-Hoc Recalibration

We evaluate three post-hoc recalibration techniques to improve the uncertainty estimates of world models without requiring retraining:

1. Temperature Scaling [4]: A single scalar parameter T is learned on a validation set by minimizing the Negative Log-Likelihood (NLL). The calibrated probability is given by $p' = \sigma(z/T)$, where z is the logit and σ is the sigmoid or softmax function.
2. Conformal Prediction [9]: Provides a distribution-free theoretical guarantee that the true outcome falls within a predicted set with a user-specified probability $1 - \alpha$.
3. Histogram Binning: A non-parametric baseline that aligns probabilities within predefined bins to match their empirical frequencies.

3.5. Downstream Planning Evaluation

To understand the practical impact of calibration, we integrate the world model into a Model Predictive Control (MPC) planning framework. The planner minimizes a cost function $J(a)$:

$$J(a) = \mathbb{E}[C(a)] + \lambda \cdot U(a), \quad (3)$$

Table 1. Calibration metrics on nuScenes val.

Model	ECE ↓	MCE ↓	Brier ↓	AUROC ↑
OccWorld	[TODO: 0.XXX]	[TODO: 0.XXX]	[TODO: 0.XXX]	[TODO: 0.XXX]
Vista	[TODO: 0.XXX]	[TODO: 0.XXX]	[TODO: 0.XXX]	[TODO: 0.XXX]

where $C(a)$ represents standard planning costs (e.g., collisions, progress, comfort), and $U(a)$ is an uncertainty penalty derived from the world model’s predictive variance or entropy. We vary the risk-aversion parameter λ from 0 (uncertainty-agnostic baseline) to 5 (highly risk-averse). We evaluate the resulting planner’s collision rate, route progress, and comfort metrics.

4. Experiments

4.1. Experimental Setup

Dataset: We use the standard nuScenes dataset [1], focusing on the validation split. We curate a subset of 1500 diverse scenarios, stratified into easy, medium, and hard difficulties.

Models: We evaluate two representative architectures:

- OccWorld [16]: An autoregressive 3D occupancy world model. We generate stochastic samples via temperature sampling.
- Vista [3]: A diffusion-based video generation world model. We induce stochasticity by varying the initial noise seed.

For each scenario and model, we generate $N = 20$ stochastic samples to estimate predictive uncertainty.

4.2. Overall Calibration (E1)

Our baseline evaluation reveals that both occupancy and video world models are significantly overconfident. As shown in Table ??, OccWorld and Vista yield ECE values of [TODO: 0.12] and [TODO: 0.15], respectively. The reliability diagrams (Figure ??) further illustrate this trend, showing that predictions with $> 90\%$ confidence often exhibit empirical accuracies below [TODO: 70%].

4.3. Spatial-Temporal Analysis (E2-E4)

Spatial Degradation: Calibration error increases dramatically with distance from the ego vehicle. Table ?? demonstrates that for OccWorld, the ECE in the $0 - 10m$ bin is [TODO: 0.05], but degrades to [TODO: 0.18] in the $50 - 100m$ bin.

Temporal Degradation: Figure ?? shows the progression of ECE over the prediction horizon. At 0.5s, models remain relatively well-calibrated, but by 3.0s, accumulated autoregressive errors and diffusion drift cause the ECE to double.

Scenario Difficulty and Dynamics: Hard scenarios exhibit significantly worse calibration than easy scenarios. Furthermore, uncertainty is particularly poorly calibrated for dynamic objects compared to static backgrounds, suggesting models struggle to properly quantify uncertainty in complex, highly dynamic interactions.

4.4. Qualitative Driving Anecdotes & Failure Modes

To understand the practical failure modes of driving world models beyond aggregate metrics, we mine and analyze specific real-world driving anecdotes from nuScenes (Figure 2):

Case 1: Blind-Zone Pedestrian Emergence (False-Negative Hazard): When a pedestrian steps into a crosswalk from behind a parked high-profile truck, the world model overconfidently predicts free space ($p < 0.15$) in the unobserved blind zone instead of expressing high epistemic uncertainty. For downstream planners, this false certainty suppresses braking, directly inducing safety-critical collision risk.

Case 2: Unprotected Left Turn (Mode Collapse): Navigating an intersection with oncoming traffic presents a bifurcating decision boundary. Rather than capturing a bimodal distribution across stochastic rollouts, the model exhibits mode collapse, predicting with unjustified certainty that oncoming agents will yield.

Case 3: High-Speed Lateral Cut-In (Dynamic Lag): When an adjacent vehicle abruptly swerves into the ego buffer, the model’s confidence exhibits a severe temporal lag, underestimating lateral displacement during transition timesteps ($t \approx 1.5s$).

Case 4: Long-Horizon Hallucination (Autoregressive Drift): At horizons $t \geq 2.5s$, compounding discrete token errors cause phantom obstacles to materialize on clear road surfaces while genuine obstacles fade, explaining the dramatic temporal calibration decay observed in Section 4.

Front-Camera Behavioral Rollouts: In addition to top-down grid representations, Figure 3 visualizes the driver’s front-camera perspective across sequential rollouts ($t = 0.5s \rightarrow 3.0s$). Projecting predictive uncertainty directly onto the physical road surface and dynamic actors reveals that world models often exhibit severe epistemic blind spots during dynamic cut-ins, failing to elevate predictive variance until after physical buffer violations occur.

4.5. Stochastic Sample Scale & Environmental Stress Tests

Stochastic Sample Efficiency ($N \in \{1, 3, 5, 10, 20\}$): An essential question for real-time deployment is how many forward passes are required for calibrated confi-

dence. Table ?? and Figure 4 demonstrate the latency-accuracy Pareto frontier. While greedy deterministic inference ($N = 1$) suffers from extreme overconfidence ($ECE = [TODO: 0.24]$), calibration error decreases sharply up to $N = 10$ ($ECE = [TODO: 0.11]$), beyond which diminishing returns set in. This confirms that $N = 10$ provides an optimal operating point for onboard vehicle compute.

Environmental Domain Shift: We stress-test calibration under adverse environmental conditions (Table ??). When transitioning from clear daytime driving to nighttime and rainy conditions, ECE increases by $[TODO: +48\%]$ and $[TODO: +62\%]$, respectively. Specular puddles and reduced camera dynamic range introduce significant perceptual noise that uncalibrated world models fail to reflect in their confidence scores.

4.6. Recalibration (E5)

We evaluate post-hoc recalibration methods in Table ?. Standard global temperature scaling reduces ECE by over $[TODO: 50\%]$. However, our proposed Spatio-Temporal Temperature Scaling $T(d, t)$ achieves the lowest calibration error across all configurations ($[TODO: 0.041]$), demonstrating the critical value of distance- and horizon-conditioned corrections. Furthermore, conformal prediction successfully provides distribution-free marginal coverage guarantees (Table ??), establishing rigorous safety envelopes for mission-critical maneuvers.

4.7. Downstream Planning (E6)

To validate the utility of calibrated uncertainty, we evaluate the MPC planner. Table ?? demonstrates that integrating calibrated uncertainty ($\lambda > 0$) reduces the collision rate by $[TODO: 30\%]$ compared to the baseline planner ($\lambda = 0$). Furthermore, Figure ?? illustrates the safety-progress trade-off: highly risk-averse settings ($\lambda = 5$) eliminate collisions but result in overly conservative driving, highlighting the importance of tuning the uncertainty penalty based on properly calibrated confidence estimates.

5. Conclusion

In this paper, we introduced CalibraDrive, the first benchmark dedicated to evaluating predictive uncertainty and calibration in driving world models. Our comprehensive analysis of representative occupancy and video models on the nuScenes dataset reveals a pervasive issue: modern generative driving models are significantly overconfident. This miscalibration worsens over longer prediction horizons and in hard driving scenarios, potentially jeopardizing downstream safety.

Figure 2. Qualitative real-world driving anecdotes mined by CalibraDrive. We highlight four archetypal corner cases: (Case 1) Blind-Zone Pedestrian Emergence, where models overconfidently predict free space in occlusions; (Case 2) Unprotected Left Turn with oncoming traffic, showing mode collapse; (Case 3) High-Speed Lateral Cut-In, revealing dynamic calibration lag; and (Case 4) Long-Horizon Hallucination at $t = 3.0$ s. Columns depict ground truth context, model predicted confidence, predictive uncertainty, and safety-critical failure regions.

Figure 3. Front-camera driver perspective and behavioral dynamics across prediction horizons ($t = 0.5$ s to 3.0 s). Row 1 shows the physical road environment and dynamic actor movements (cut-in vehicle and roadside pedestrian). Row 2 projects the model’s predictive uncertainty heatmap directly onto the camera view. Row 3 depicts synchronized BEV occupancy and ego planned path.

Figure 4. Stochastic sample count ablation. (a) Calibration error (ECE) as a function of sample scale N . (b) Latency-accuracy Pareto frontier showing optimal convergence around $N = 10$ samples.

Crucially, we show that simple post-hoc recalibration techniques, such as temperature scaling, effectively mitigate this overconfidence and are easy to adopt. Furthermore, integrating these calibrated uncertainty estimates into downstream planning significantly reduces collision rates, highlighting the practical necessity of uncertainty-aware world modeling.

Future work will explore online, dynamic calibration methods and the integration of calibration objectives directly during the training phase of driving world models. We hope CalibraDrive will serve as a standard evaluation axis, encouraging the development of not just highly realistic, but highly reliable world models for autonomous driving.

Code and evaluation scripts will be made publicly available.

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